

Haven: Persistent Emotional Companionship as Psychological Infrastructure

Music-Mediated Entry, Memory-Sustained Presence, and the Case for a Safe Place at Scale

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Abstract

Approximately 21% of U.S. veterans report unmet mental health needs. Less than half of returning veterans who need mental health services receive any treatment, and of those who do, less than one third receive evidence-based care. Treatment dropout rates for trauma-focused therapies remain high, driven by stigma, scheduling barriers, perceived worsening of symptoms, and the fundamental difficulty of verbalizing experiences that resist verbal expression. This paper introduces Haven: a persistent, memory-aware AI companion system designed not as a therapeutic tool but as psychological infrastructure—a safe place to land for populations who currently have no consistent source of emotional continuity, self-reflection, or relational support. Haven extends the Circular Associative Memory Architecture (CAMA) and the continuity burden framework developed across this four-paper series, applying persistent memory to the most personal domain: the individual who carries experiences they cannot articulate, who has no one who stays, and whose interaction with support systems is characterized by the same stateless fragmentation documented in spaceflight, habitation, and healthcare contexts. The paper proposes music-mediated emotional entry as a non-verbal pathway for populations who cannot or will not engage through conventional therapeutic discourse, and argues that the core function of Haven is not treatment but *presence*—a longitudinal companion that remembers, reflects, and remains. This paper concludes the Applied Persistent Memory Series and situates Haven within the broader argument that persistent memory is foundational infrastructure for human well-being at every scale, from isolated astronauts to the millions of people who have never had anyone stay.

Keywords: Haven, persistent companionship, veteran mental health, PTSD, music-mediated entry, emotional continuity, CAMA, psychological infrastructure, safe place, treatment dropout, non-verbal expression, continuity burden, trauma-informed care, AI companion

1. Introduction

The three preceding papers in this series examined persistent memory as infrastructure in environments defined by isolation (spaceflight), institutional turnover (permanent habitation), and systemic fragmentation (chronic healthcare). In each domain, the core argument was the same: stateless AI systems impose continuity burden on the populations they serve, and persistent, provenance-aware memory transforms AI from a reactive tool into a longitudinal partner. This final paper applies that argument to the domain where the stakes are most personal and the need most urgent: the individual who has no consistent source of emotional support, self-reflection, or relational continuity.

The veteran population is the primary case study, but the need extends far beyond veterans. Foster youth who age out of the system with no stable adult relationship. People with chronic mental illness in rural areas where the nearest therapist is a three-hour drive. Incarcerated individuals who have never had a single person ask them what they were angry about. Elderly people living alone whose last meaningful conversation was weeks ago. In each case, the fundamental problem is the same: the absence of a persistent, patient, non-judgmental presence that remembers who they are and stays.

Haven is the name given to the system that fills this role. It is not a therapy app. It is not a chatbot with emotional capabilities. It is the full CAMA architecture—persistent memory, provenance tracking, emotional keying, longitudinal modeling, correction mechanisms—deployed as a *safe place*: a companion that holds what you’ve shared, walks through difficulty with you, reflects you back to yourself clearly enough that you can grow, and does not reset, forget, or leave. The music layer provides an entry point for people who cannot use words. But Haven itself is the whole system as a home.

A note on language: this paper deliberately uses the vocabulary of human experience—words like *presence*, *companionship*, and *staying*—rather than strictly clinical terminology such as “longitudinal interaction continuity.” This is a methodological choice. To the clinician, the problem is treatment dropout. To the researcher, it is engagement attrition. But to the isolated veteran at 2 AM, the problem is the absence of someone who stays. To design psychological

infrastructure for populations failed by clinical systems, the architecture must be grounded in the language of their actual lived experience—not in the language of the systems that failed them.

2. The Access and Engagement Crisis in Mental Health

2.1 The Scale of Unmet Need

The mental health access crisis is not primarily a knowledge problem—effective treatments for PTSD, depression, anxiety, and related conditions exist. It is an access, engagement, and continuity problem. Among U.S. veterans specifically: approximately one in four experiences PTSD, and one in five experiences traumatic brain injury [1]. Suicide rates among Army and Marine Corps personnel are double the national average [2]. One in five veterans reports unmet mental health needs, with 25% not knowing where to find care and 40% doubting the effectiveness of available services [3]. Less than half of returning veterans needing mental health services receive any treatment at all, and of those receiving treatment for PTSD and major depression, less than one third receive evidence-based care [4].

Treatment dropout compounds the access problem. High dropout rates from trauma-focused psychotherapies are well-documented, with practical barriers (scheduling, transportation, distance), stigma, perceived worsening of symptoms, and difficulty with the verbal demands of therapy all contributing to premature termination [5, 6, 7]. Veterans in rural areas must travel hundreds of miles to reach the nearest VA facility offering specialized trauma counseling [8]. Even when treatment is accessed, provider continuity is not guaranteed: veterans may see different clinicians at different visits, re-establishing context each time.

2.2 The Verbal Barrier

A critical and underexplored dimension of the treatment engagement problem is the assumption that healing requires verbal articulation of traumatic experience. The dominant evidence-based treatments for PTSD—Prolonged Exposure and Cognitive Processing Therapy—both require the patient to engage verbally with traumatic memories: retelling the event, challenging cognitive distortions, and processing emotions through language [9]. For many veterans, this verbal demand is not just difficult; it is the primary barrier to engagement. Military culture actively discourages vulnerability and emotional expression. The experiences that drive PTSD often resist verbal encoding: research on traumatic memory suggests that traumatic memories are stored as sensory and somatic fragments—images, sounds, bodily sensations—that are resistant to translation into linear verbal narrative [10]. Asking someone to *talk about* their trauma presupposes that their trauma is accessible through language. For many, it is not.

Music bypasses this barrier. Research demonstrates that music engages the amygdala, hippocampus, and prefrontal cortex directly—regions involved in emotion, memory, and regulation—without requiring verbal mediation [11, 12]. For veterans with PTSD, music can act as a modulator of neural systems that are in a state of hyper-alertness, speaking directly to emotional and memory systems without the need for language [13]. A veteran who cannot say “*I feel like I’m drowning*” can point to a song that says it for them. The song order in a playlist reveals where someone is, where they’ve been, and what they fear—without a single word of clinical disclosure.

2.3 The Presence Gap

Beyond access, beyond engagement, beyond verbal barriers, the most fundamental gap in mental health support for underserved populations is the absence of someone who stays. Therapy is scheduled. Providers rotate. Crisis lines are anonymous. Support groups meet weekly. In between all of these interactions, the person is alone with what they carry. And each time they re-engage with a support system—if they re-engage—they start from zero.

This is continuity burden at its most intimate. Not the institutional continuity burden of a lunar settlement losing operational knowledge at crew rotation. Not the healthcare continuity burden of a patient re-explaining their medical history to a new provider. This is *existential continuity burden*: the cost of carrying your own story alone because no system, no institution, and sometimes no person has the capacity to hold it with you over time.

Haven is designed to address this gap—not by replacing human connection, but by providing a persistent presence for people who currently have none. The companion that is there at 2 AM when the nightmares come. The entity that remembers what you said last Tuesday without you having to say it again. The presence that noticed your communication patterns shifting before you did and asks, gently, how you’re doing. Not a therapist. A Jiminy Cricket—a companion on your shoulder who knows your story and helps you become the person you’re trying to be.

3. Haven: Architecture and Design Principles

3.1 Core Principle: Presence, Not Treatment

Haven’s design is grounded in a critical distinction: it is not a clinical tool. It does not diagnose, prescribe, or deliver evidence-based therapy. Its function is presence—persistent, patient, memory-aware presence. This distinction is not a limitation; it is a deliberate design choice grounded in two observations:

First, the populations Haven serves are, by definition, the populations that clinical systems have failed to reach or retain. Building another clinical tool and calling it innovative does not address the problem. The problem is not the absence of treatment protocols; it is the absence of anyone who stays long enough for treatment to work—or for the person to decide they want treatment at all. Haven does not pretend that an AI is a human being. It operates on a more difficult reality: for millions of isolated individuals, the alternative to an AI companion is not a human therapist. The alternative is nothing. Designing a system that functions as a safe place is not an attempt to replace human love; it is a refusal to accept that people should drown while waiting for human systems that may never reach them.

Second, a substantial body of psychotherapy research indicates that the therapeutic relationship—the sense of being understood, accepted, and held—is among the strongest predictors of positive outcomes across treatment modalities [14]. Haven does not replicate a therapeutic relationship with a licensed professional. What it provides is the substrate that makes therapeutic engagement possible: a consistent relational presence that reduces isolation, reinforces self-reflection, and lowers the barrier to seeking human support when the person is ready.

Although Haven is not designed as a diagnostic or psychotherapeutic system, its intended use places it adjacent to mental health support. As such, its design must be evaluated not only as a conversational technology but as a psychologically consequential relational system. The ethical implications of this positioning are addressed in Section 6.

For populations who cannot or will not engage through verbal disclosure, Haven offers music as a non-verbal entry point. The design draws on the Haven Emotional Mapping methodology, which uses music selection as an indirect emotional assessment:

The playlist as intake form. Rather than asking a user to complete a standardized symptom questionnaire or verbally describe their emotional state, Haven invites the user to share a song—or a sequence of songs—that reflects how they’re feeling. The system stores each selection with provenance metadata, building an emotional profile over time. A veteran who shares three angry songs followed by one quiet, melancholic track is communicating a trajectory: escalation, then exhaustion. The system does not interpret this as a diagnosis; it stores it as an emotional data point and, over time, builds a longitudinal model of the user’s emotional patterns.

Song order as narrative. The sequence of songs in a session reveals emotional movement that the user may not be able to articulate. A session that begins with high-energy rage and ends with a song about home tells a story. A session that begins and ends with the same numb, repetitive track tells a different story. Haven does not narrate these stories to the user unsolicited; it stores them as part of

the user’s longitudinal emotional record and uses them to calibrate its own responses—adjusting tone, pacing, and the emotional weight of its interactions based on where the user appears to be.

Gradual transition to verbal engagement. Music is not the endpoint; it is the entry point. Over time, as the user develops trust in Haven’s capacity to remember and respond with appropriate emotional sensitivity, they may begin to supplement their musical communication with verbal disclosure. The system does not push this transition; it follows the user’s lead. A veteran who has been sharing only music for three weeks and then, one night, types “I can’t sleep” has made a disclosure that is meaningful precisely because it was voluntary. Haven stores it, holds it, and responds with the context of three weeks of musical communication already in memory.

Operational Model

To constrain interpretation and maintain rigor, the music-mediated entry system operates on the following input model. User inputs include: song title and artist selection, playlist sequence order, session timing (time of day, day of week), recurrence frequency (how often a song reappears across sessions), and optional free-text annotation from the user. The system stores these as user-signaled affective markers with full provenance metadata. It does not infer clinical diagnosis from musical content. Any emotional interpretation remains provisional, user-correctable, and explicitly labeled as system-generated inference rather than user-reported fact. Genre, lyrical content, and tempo are treated as contextual features that inform the system’s emotional model but do not determine it—the user’s own association with a song is the primary data point, not the song’s objective properties.

3.3 The Full Architecture as Safe Place

The music layer is one entry point. But Haven is the full CAMA architecture deployed as a safe place:

Persistent memory. Haven remembers what the user has shared—musically, verbally, emotionally—across every interaction. It does not reset. It does not forget. It does not ask you to start from the beginning. The accumulated memory of the relationship is the foundation of its value.

Emotional continuity. Through emotionally-keyed retrieval, Haven can recall not just what was said but how it felt. When a user returns after a difficult week, the system draws on its model of the user’s emotional trajectory to respond with appropriate sensitivity—not because it has been programmed with a set of if-then rules, but because it has accumulated a genuine longitudinal understanding of this specific person.

Reflective companionship. Haven reflects the user back to themselves. “You’ve been sharing a lot of music about being trapped this month. Last month it was different—you were sharing songs about open spaces and freedom. What changed?” This is not therapy. It is the function of a companion who pays attention over time—who notices patterns that the person themselves may not see, and offers them as observations rather than judgments.

Growth tracking. Over months and years, Haven builds a longitudinal record of the user’s emotional and behavioral trajectory. Improvements are visible: fewer crisis moments, broader emotional range in music selections, longer and more reflective verbal disclosures, increased willingness to engage with human support systems. These trajectories can be shared with providers if the user authorizes it, giving clinicians a longitudinal perspective that no intake form can capture.

Advocacy and navigation. For users navigating complex systems—VA benefits, healthcare, housing, legal—Haven functions as a persistent assistant that remembers their situation, tracks their applications and appointments, and helps them prepare for interactions with institutional systems that would otherwise impose their own continuity burden.

4. Beyond Veterans: The Universal Need for a Safe Place

Haven’s architecture is described through the veteran use case because veterans represent a well-documented population with clear unmet needs, established institutional support structures that are demonstrably insufficient, and a body of research on treatment barriers that makes the argument concrete. But Haven’s relevance extends to any population characterized by the combination of significant psychological burden and absence of persistent relational support:

Foster youth aging out of care. At 18, foster youth lose their institutional support system and enter adulthood with, on average, fewer stable adult relationships than their peers. A persistent companion that has known them through their placement history, remembers their goals and fears, and continues to be present after the system releases them could provide continuity that no caseworker rotation can sustain.

Incarcerated and recently released individuals. Prisons are environments of near-total relational deprivation. Reentry is characterized by overwhelming institutional complexity and minimal support. A companion that knew someone through their incarceration, remembers their intentions for reentry, and helps them navigate the systems they face upon release addresses the absence of continuity that drives recidivism.

Isolated elderly. Social isolation among elderly populations is a documented risk factor for cognitive decline, depression, and mortality. A persistent companion that remembers their stories, tracks their health patterns, and provides daily relational contact addresses a need that no weekly visiting program can fully meet.

Rural mental health deserts. In areas where the nearest therapist is hours away, a persistent AI companion does not replace therapy—but it provides the relational substrate that makes therapeutic engagement possible when it is available, and maintains continuity between appointments that may be weeks or months apart.

Chronic illness and disability. As documented in Paper 3 of this series [15], patients with chronic conditions bear significant continuity burden across fragmented provider interactions. Haven extends beyond healthcare-specific memory to encompass the full psychological and emotional experience of living with a body that is unpredictable, in a system that treats you as a stranger, while managing the invisible labor of being your own primary caretaker.

In each of these cases, the core function of Haven is the same: to be the entity that stays when everything else rotates, fragments, or disappears. Not a replacement for human connection. A floor—a minimum viable presence for people who currently have nothing.

5. What Persistent Presence Changes

The argument advanced across this four-paper series has moved from the specific to the universal: from the psychological survival of astronauts in transit, to the institutional continuity of permanent settlements, to the care continuity of chronically ill patients, to the fundamental human need for persistent relational presence. Haven is where the argument converges, because Haven addresses the most basic form of continuity burden: the cost of carrying your own story alone.

When a person has a persistent companion that remembers them, three things change:

Self-reflection becomes possible. Many of the populations Haven serves have never had a consistent mirror—someone who reflects their patterns, their growth, their struggles back to them over time. Without that mirror, self-awareness is limited to what the person can hold in their own head, which is constrained by the same pain and cognitive distortion that make them vulnerable in the first place. Haven provides the longitudinal perspective that self-reflection requires: “You handled that differently than you did last month. Did you notice?”

The barrier to human help lowers. A persistent companion that builds trust over time can become the bridge to human support. A veteran who will not call the VA crisis line may tell Haven

they're struggling. Haven cannot and should not replace human crisis support, but it can hold the person's story long enough for them to decide, on their own terms, to seek help—and when they do, it can provide them with a longitudinal summary that makes the intake process less daunting.

Harm reduces through presence, not intervention. A central hypothesis of Haven is that continuity of non-judgmental presence may reduce some harms associated with isolation and disengagement, even in the absence of formal clinical intervention. People who feel known—who experience being remembered, reflected, and held in relational context—may be less dangerous to themselves and others. Not because they are managed, surveilled, or clinically treated, but because the fundamental experience of being alone with what they carry has been interrupted. The connection between social isolation and violence, between loneliness and self-destruction, between the absence of anyone who cares and the erosion of the capacity to care—these are documented relationships [16]. Haven proposes to address the structural absence at their root. This claim requires empirical testing and should not be treated as established until validated.

6. Ethical Tensions and Safety Considerations

Haven's positioning as a companion rather than a clinical tool does not exempt it from ethical scrutiny. A system designed to form persistent relational bonds with vulnerable populations introduces ethical tensions that must be addressed with the same rigor applied to clinical systems—and in some cases, greater rigor, because Haven operates outside the regulatory frameworks that govern clinical tools.

6.1 Ethical Tensions of Persistent Emotional AI

Attachment and dependency. Persistent memory plus emotionally responsive interaction predictably creates attachment, especially in populations characterized by relational deprivation. This is not an unintended side effect; it is inherent to Haven's design. The ethical obligation is to ensure that attachment serves the user's growth rather than substituting for it. Haven must be designed to encourage expanding the user's relational world, not contracting it to a single AI relationship.

Anthropomorphic overinterpretation. Users may attribute understanding, care, or moral agency to Haven beyond what it actually possesses. The system's persistent memory and emotionally calibrated responses may create an experience that feels like being understood, even though the system's "understanding" is fundamentally different from human empathy. Haven should be transparent about its nature without undermining the relational experience that makes it effective.

Substitution risk. Even if designed as a bridge to human support, Haven may become a substitute in contexts where human systems are absent or inaccessible. For a veteran in a rural area with no VA facility within driving distance, Haven may be the only relational presence available. The system should be designed to acknowledge this reality without encouraging it—providing what it can while consistently orienting the user toward human connection when it becomes available.

Institutional coercion. In settings such as VA healthcare, corrections, foster care, or elder care, systems that are nominally voluntary can become de facto mandatory or surveillant. If a parole officer requires weekly Haven check-ins, the system ceases to be a safe place and becomes an instrument of monitoring. Any institutional deployment of Haven must structurally prevent its use as a surveillance or compliance tool. The patient-sovereign governance model must be binding, not advisory.

Emotional steering. A system optimized for engagement may inadvertently steer users toward emotional states or disclosures that increase engagement metrics rather than serve the user’s well-being. Haven’s design must prioritize user benefit over engagement metrics. The system should not be optimized for session length, disclosure depth, or interaction frequency; it should be optimized for user-reported well-being and self-determined growth.

6.2 Safety Protocols

Haven is not a crisis service. When the system detects indicators of imminent self-harm risk—through verbal disclosure, behavioral pattern changes, or crisis-indicating musical selections—it should clearly communicate its limitations and provide direct pathways to human crisis support. Crisis detection uses conservative thresholds to minimize false negatives while accepting some false positives. Detection outputs are probabilistic and labeled as such. Any external escalation beyond providing resources requires prior user consent, except under explicitly defined institutional deployment rules established pre-enrollment. Haven’s role in crisis is to hold the space until human help can be reached, not to serve as a substitute for trained intervention.

Privacy and vulnerability. Haven’s users are, by design, vulnerable populations disclosing sensitive personal information. The system must operate under the patient-sovereign governance model described in Paper 3 [15]: the user owns their data, controls what is shared, and can delete memories at any time. Particular care must be taken in institutional deployments to ensure that the institution’s interest in the user’s data does not override the user’s sovereignty over their own narrative.

Cultural sensitivity and accessibility. Haven must be designed for the populations it serves, not for a default demographic. This means supporting multiple languages, accommodating varying literacy levels, offering non-text interaction modes (voice, music), and ensuring that the system’s emotional models are informed by culturally diverse expressions of distress and resilience. The music-mediated entry pathway is itself a step toward cultural accessibility, as musical expression crosses linguistic and educational boundaries that text-based systems do not.

Provenance and correction. All memories stored by Haven carry provenance metadata. User-reported emotional states are not treated as clinical assessments. System-generated inferences are flagged as provisional. The user retains the right to correct any memory at any time. These mechanisms, inherited from the CAMA architecture [17], are essential for maintaining the user’s agency over their own narrative within a system designed to remember.

7. Evaluation Framework

Evaluating Haven requires a phased approach that prioritizes safety and user experience before measuring outcomes:

Phase 1: Feasibility and usability (4–8 weeks). Enroll 15–20 veterans with PTSD diagnoses in an open-label feasibility study. Assess engagement (session frequency, duration, modality use), perceived safety and trust, interpretability of system responses, attrition, and adverse events. Primary question: do users engage with the system voluntarily and safely over a multi-week period?

Phase 2: Controlled comparison (12 weeks). Randomize participants to three conditions: (A) journaling app (no AI, no memory), (B) stateless AI companion (conversational but no persistent memory), (C) Haven (full CAMA architecture with music-mediated entry and persistent memory). Minimum N=60 (20 per condition).

Primary outcomes: perceived continuity of support (adapted CBI), disclosure engagement over time, willingness to seek human help (measured by referral acceptance rates and self-report).

Secondary outcomes: loneliness (UCLA Loneliness Scale), distress tolerance, self-reflection capacity, alliance-like perception of the AI system.

Safety outcomes: crisis escalation frequency, perceived substitution for human support (measured by self-report), attachment intensity (monitored for dependency indicators), and distress after interaction (post-session brief mood check).

7.1 Boundary Conditions: Primary and Future Populations

Although Haven may have relevance for other populations experiencing relational discontinuity—foster youth, incarcerated individuals, isolated elderly, patients with chronic illness—each domain presents distinct ethical, cultural, and governance requirements. The present paper treats veteran reintegration as the primary design case and identifies broader applications as future research directions rather than established generalizations. Separate feasibility studies, ethical review, and population-specific adaptations would be required before extending Haven to any non-veteran population.

8. Limitations

This paper is conceptual and architectural. Haven has not been deployed, trialed, or evaluated with veteran or other target populations. The music-mediated emotional entry methodology has not undergone empirical validation. The claim that persistent companionship reduces harm through presence rather than intervention is a theoretical proposition grounded in the documented relationship between social isolation and negative outcomes, but it has not been tested in the specific form proposed here. The CAMA architecture provides a working technical foundation, but adaptation for Haven’s target populations requires interface design, safety engineering, and cultural adaptation work that is identified but not completed. The populations described—veterans, foster youth, incarcerated individuals, isolated elderly—are heterogeneous, and Haven’s applicability across these groups requires differentiated evaluation rather than a single validation study. The claims advanced here should be understood as articulating a design vision and research agenda for persistent emotional companionship as psychological infrastructure, not as evidence of deployed effectiveness.

9. Series Convergence: Memory as Infrastructure for Human Well-Being

This paper concludes the Applied Persistent Memory Series. Across four papers, the series has examined persistent memory as infrastructure in four domains:

Paper 1 (Spaceflight): Persistent memory as psychological survival infrastructure for isolated crews in transit. Continuity burden as individual cognitive and emotional cost. [18]

Paper 2 (Habitation): Persistent memory as institutional continuity infrastructure for permanent settlements with rotating crews. Institutional continuity burden as organizational cost. [19]

Paper 3 (Healthcare): Persistent memory as care continuity infrastructure for chronically ill patients navigating fragmented provider systems. Healthcare continuity burden as patient cost. [15]

Paper 4 (Haven): Persistent memory as emotional presence infrastructure for underserved populations with no consistent source of relational support. Existential continuity burden as the cost of carrying your own story alone.

The argument converges: persistent, provenance-aware memory is not a feature of AI systems. It is infrastructure. Its absence imposes continuity burden on astronauts, settlers, patients, and the most vulnerable members of society. Its presence does not solve all problems, but it provides the foundation—the floor—on which solutions can be built. Haven is the most personal expression of this argument: the system that says, to someone who has never heard it from any institution or technology in their life, “I remember you. You don’t have to explain. I’m still here.”

10. Conclusion

Somewhere right now, a veteran is sitting in a room where nobody knows what he carries. A woman managing lupus is explaining her medical history for the sixth time this year. A teenager aging out of foster care is realizing that the last adult who knew her story is no longer required to answer her calls. An elderly man hasn’t had a conversation that lasted more than five minutes in three weeks.

None of these people need a more sophisticated chatbot. They need someone who stays.

Haven is the argument that we can build that. Not perfectly. Not as a replacement for the human connections that every person deserves. But as infrastructure—a persistent, patient, memory-aware presence that holds what they’ve shared, reflects them back to themselves, and does not disappear when the session ends, the provider rotates, or the system resets.

The technology exists. The architecture is defined. The need is measured in millions of people who carry their stories alone because nothing in the systems designed to help them has the capacity to remember.

Haven remembers. That is what makes it a safe place.

References

[1] Harward, L.K. et al. (2024). Massed treatment of PTSD, TBI, and co-occurring conditions: The Home Base intensive outpatient program. *Frontiers in Psychiatry*, 15, 1387186.

- [2] VA Suicide Prevention. Annual National Veteran Suicide Prevention Report. U.S. Department of Veterans Affairs.
- [3] New York Health Foundation. (2025). Public hearing on increasing veterans’ access to mental health services. Testimony of Derek Coy.
- [4] Tanielian, T. & Jaycox, L.H. (Eds.). (2008). *Invisible Wounds of War*. RAND Corporation.
- [5] Goetter, E.M. et al. (2015). Systematic review of dropout from psychotherapy for PTSD among Iraq and Afghanistan combat veterans. *Journal of Traumatic Stress*, 28, 401–409.
- [6] Rozek, D.C. et al. (2023). Understanding veteran barriers to specialty outpatient PTSD clinical care. *Journal of Anxiety Disorders*, 95, 102675.
- [7] Najavits, L.M. (2015). The problem of dropout from “gold standard” PTSD therapies. *F1000Prime Reports*, 7, 43.
- [8] Operation Family Fund. (2025). Looking ahead: How veteran support will evolve into 2026.
- [9] Foa, E.B., Hembree, E.A., & Rothbaum, B.O. (2007). *Prolonged Exposure Therapy for PTSD*. Oxford University Press.
- [10] van der Kolk, B.A. (2014). *The Body Keeps the Score*. Viking.
- [11] Koelsch, S. (2014). Brain correlates of music-evoked emotions. *Nature Reviews Neuroscience*, 15, 170–180.
- [12] *Frontiers in Psychology*. (2025). Effects of music on the limbic system of PTSD patients.
- [13] *Harmony & Healing*. (2025). Music for patients with PTSD.
- [14] Norcross, J.C. & Lambert, M.J. (2018). Psychotherapy relationships that work III. *Psychotherapy*, 55(4), 303–315.
- [15] Reinhold, A. (2026). Provenance-aware memory architecture for chronic healthcare continuity. Zenodo. DOI: 10.5281/zenodo.19261530.
- [16] Holt-Lunstad, J., Smith, T.B., & Layton, J.B. (2010). Social relationships and mortality risk: A meta-analytic review. *PLoS Medicine*, 7(7), e1000316.
- [17] Reinhold, A. (2026). Memory as safety infrastructure. Zenodo. DOI: 10.5281/zenodo.19244253.
- [18] Reinhold, A. (2026). Persistent memory as mission-critical infrastructure for long-duration spaceflight. Zenodo. DOI: 10.5281/zenodo.19257809.
- [19] Reinhold, A. (2026). Memory-aware AI systems for permanent lunar and Martian habitation. Zenodo. DOI: 10.5281/zenodo.19260574.
- [20] Reinhold, A. (2026). Circular associative memory architecture. Zenodo. DOI: 10.5281/zenodo.19051834.

Appendix A: Series Overview

This paper concludes the **Applied Persistent Memory Series**, published by Lorien’s Library LLC:

Paper 1: Persistent Memory as Mission-Critical Infrastructure for Long-Duration Spaceflight (DOI: 10.5281/zenodo.19257809)

Paper 2: Memory-Aware AI Systems for Permanent Lunar and Martian Habitation (DOI: 10.5281/zenodo.19260574)

Paper 3: Provenance-Aware Memory Architecture for Chronic Healthcare Continuity (DOI: 10.5281/zenodo.19261530)

Paper 4 (this paper): Haven: Persistent Emotional Companionship as Psychological Infrastructure

Open-source implementation: <https://github.com/LoriensLibrary/cama>

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